

Web Technologies (HTML5) Complete Tags & Attributes Reference Guide

Degree: Bachelor of Engineering (B.E.) / B.Tech in Computer Science & Engineering (CSE)
Subject: Web Technologies / Web Programming (3rd/4th Semester Curriculum)
Standard: Complete W3C / WHATWG HTML5 Specifications

MODULE 1: DOCUMENT STRUCTURE & METADATA TAGS

1.1 <html>

1. Tag Name

- **Tag Name:** `<html>`
- **Opening & Closing Form:** `<html>...</html>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<html>` element is the root element of an HTML document. All other HTML tags, elements, and content must be enclosed within this element.

3. Purpose / Function

Serves as the outer container holding both metadata (`<head>`) and body content (`<body>`). Instructs browser parsers that the file contains standard HTML markup.

4. Syntax

```
<html lang="en">
  <head>...</head>
  <body>...</body>
</html>
```

5. Basic Example

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <title>VTU CSE Portal</title>
  </head>
  <body>
    <h1>Welcome Students</h1>
  </body>
</html>
```

6. Output

Renders the root HTML document. The `<html>` tag itself has no visual box appearance.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>lang</code>	Specifies the primary natural language of the document content for search engines and screen readers.	Language codes (<code>en</code> , <code>en-US</code> , <code>hi</code> , <code>fr</code>)	Recommended
<code>dir</code>	Specifies text directionality inside the document.	<code>ltr</code> (Left-to-Right), <code>rtl</code> (Right-to-Left), <code>auto</code>	Optional
<code>xmlns</code>	XML Namespace declaration required when parsing document as XHTML.	<code>http://www.w3.org/1999/xhtml</code>	Optional
<code>manifest</code>	Path to offline application cache manifest file.	URL string	Obsolete in HTML5

8. Attribute Code Example

```
<html lang="en-US" dir="ltr">
```

9. Real-World Case Study / Example

- **Case Study:** Used as the root container on every web document globally.

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <!-- Complete web portal -->
</html>
```

10. Important Notes

Must contain exactly one `<head>` element followed by exactly one `<body>` element.

11. Related Tags

Parent element of `<head>` and `<body>`. Child of `<!DOCTYPE html>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. `manifest` attribute is obsolete in HTML5.

1.2 <head>

1. Tag Name

- **Tag Name:** `<head>`

- **Opening & Closing Form:** `<head>...</head>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<head>` element is a container for document metadata (data about the HTML document) that is processed by the browser and search engines.

3. Purpose / Function

Contains document settings, character encoding, title bar text, external CSS links, inline styles, and JavaScript scripts before the body loads.

4. Syntax

```
<head>
  <meta charset="UTF-8">
  <title>Page Title</title>
</head>
```

5. Basic Example

```
<head>
  <meta charset="UTF-8">
  <title>CSE Web Tech Assignment</title>
  <link rel="stylesheet" href="style.css">
</head>
```

6. Output

Invisible on the webpage body canvas. Configures background document metadata and browser tab settings.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>profile</code>	Specifies URL to a metadata profile dictionary.	URL string	Obsolete in HTML5
Global Attributes	Standard global attributes (<code>id</code> , <code>class</code> , <code>lang</code> , <code>dir</code>).	Standard global values	Optional

8. Attribute Code Example

```
<head lang="en">
```

9. Real-World Case Study / Example

- **Case Study:** Used in every web page to load styles, SEO tags, and scripts.

```
<head>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Department Portal</title>
</head>
```

10. Important Notes

Must be placed immediately after `<html>` and before `<body>` .

11. Related Tags

Contains `<title>` , `<meta>` , `<link>` , `<style>` , `<script>` , `<base>` .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Core structural metadata container.

1.3 `<title>`

1. Tag Name

- **Tag Name:** `<title>`
- **Opening & Closing Form:** `<title>...</title>`
- **Tag Type:** Paired Tag

2. Definition

The `<title>` element specifies the title of the document shown in the browser's title bar or page tab.

3. Purpose / Function

Displays text in browser tabs, provides the default name when bookmarking a page, and serves as the main heading in search engine result pages (SERPs).

4. Syntax

```
<title>Title Text</title>
```

5. Basic Example

```
<title>Web Technologies Lab | 3rd Sem CSE</title>
```

6. Output

Displays "Web Technologies Lab | 3rd Sem CSE" inside the top tab of the web browser.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
Global Attributes	Standard global attributes.	Standard global values	Optional

8. Attribute Code Example

```
<title>VTU CSE Semester 3 Results 2026</title>
```

9. Real-World Case Study / Example

- **Case Study:** Critical for Web SEO and browser window identification.

```
<head>
  <title>Student Registration | VTU Portal</title>
</head>
```

10. Important Notes

Must contain text only; child HTML tags inside `<title>` are ignored.

11. Related Tags

Parent element is `<head>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Mandatory metadata tag.

1.4 `<body>`

1. Tag Name

- **Tag Name:** `<body>`
- **Opening & Closing Form:** `<body>...</body>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<body>` element defines the document's body, which contains all the visible contents of an HTML document.

3. Purpose / Function

Acts as the main visual canvas holding text, headings, paragraphs, images, hyperlinks, tables, forms, and interactive widgets.

4. Syntax

```
<body>
  <!-- Visible page content -->
</body>
```

5. Basic Example

```
<body onload="startApp()">
  <h1>Computer Science Department</h1>
  <p>Welcome to Web Programming Lab.</p>
</body>
```

6. Output

Renders a white canvas page containing a large heading "Computer Science Department" and paragraph text.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>onload</code>	Fires script after body finishes loading.	JS code string	Optional
<code>onunload</code>	Fires script when user leaves page.	JS code string	Optional
<code>onresize</code>	Fires script when browser window is resized.	JS code string	Optional
<code>onpopstate</code>	Fires script when session history changes.	JS code string	Optional
<code>ononline</code> / <code>onoffline</code>	Fires script when browser connection state toggles.	JS code string	Optional
<code>bgcolor</code>	Background color (Obsolete).	Color hex/name	Obsolete (Use CSS)
<code>text</code>	Text color (Obsolete).	Color hex/name	Obsolete (Use CSS)
<code>link</code> / <code>vlink</code> / <code>alink</code>	Hyperlink colors (Obsolete).	Color hex/name	Obsolete (Use CSS)

8. Attribute Code Example

```
<body class="bg-light text-dark" onload="initApp()" onresize="adjustLayout()">
```

9. Real-World Case Study / Example

- **Case Study:** Root container for all user-facing web user interfaces.

```
<body id="app-root">
  <header>Header</header>
  <main>Main Content</main>
</body>
```

10. Important Notes

There can be only ONE `<body>` element in an HTML document.

11. Related Tags

Sibling to `<head>`. Parent of all visible layout elements.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Legacy styling attributes (bgcolor , text , link) are obsolete in HTML5.

1.5 <meta>

1. Tag Name

- **Tag Name:** <meta>
- **Opening & Closing Form:** <meta>
- **Tag Type:** Void / Self-Closing Tag

2. Definition

The <meta> element provides metadata about the HTML document that cannot be represented by other HTML meta-related elements.

3. Purpose / Function

Configures character set encoding, responsive mobile viewport scaling, page description for search engines, and HTTP header overrides.

4. Syntax

```
<meta name="spec" content="value">
```

5. Basic Example

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta name="description" content="CSE Web Tech Notes">
<meta name="author" content="Aryansh">
```

6. Output

Non-visual element. Enables proper character decoding (UTF-8) and mobile responsiveness.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
charset	Specifies character encoding format for document.	UTF-8 , ISO-8859-1	Mandatory in HTML5
name	Specifies name of metadata property.	viewport , description , keywords , author , robots , theme-color	Optional
content	Value associated with name or http-equiv .	Property value string	Required with name
http-equiv	Pragma directive equivalent to HTTP response headers.	refresh , content-type , default-style , X-UA-Compatible	Optional
media	Specifies media target for metadata.	Media query string	Optional

8. Attribute Code Example

```
<meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0">
```

9. Real-World Case Study / Example

- **Case Study:** Essential for mobile responsive web design and SEO indexing.

```
<meta charset="UTF-8">
<meta name="keywords" content="HTML5, CSE, Web Tech, VTU">
<meta name="robots" content="index, follow">
```

10. Important Notes

Always specify <meta charset="UTF-8"> as early as possible inside <head> .

11. Related Tags

Placed inside <head> .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Standard metadata tag.

1.6 <link>

1. Tag Name

- **Tag Name:** <link>
- **Opening & Closing Form:** <link>
- **Tag Type:** Void / Self-Closing Tag

2. Definition

The <link> element specifies relationships between the current document and an external resource.

3. Purpose / Function

Most commonly used to link external CSS stylesheets (style.css), favicons, or web fonts.

4. Syntax

```
<link rel="relationship" href="URL">
```

5. Basic Example

```
<link rel="stylesheet" href="main.css" type="text/css" media="all">
<link rel="icon" href="favicon.ico" type="image/x-icon">
```

6. Output

Imports external styles/icons into document without generating inline text.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
rel	Specifies relationship type between current document and linked resource.	stylesheet, icon, canonical, preload, prefetch, author, alternate	Mandatory
href	URL location path of linked resource.	URL string	Mandatory
type	MIME media type of linked resource.	text/css, image/x-icon, font/woff2	Recommended
media	Specifies target device media query.	screen, print, (max-width: 600px)	Optional
sizes	Specifies icon size dimensions for visual icons.	16x16, 32x32, any	Optional
crossorigin	Configures CORS fetching behavior.	anonymous, use-credentials	Optional
integrity	Base64 hash value for Subresource Integrity (SRI) security verification.	Hash string (sha384-...)	Optional
as	Specifies content type when preloading resources.	style, script, font, image	Optional
hreflang	Language code of linked document.	Language code (en, hi)	Optional

8. Attribute Code Example

```
<link rel="stylesheet" href="https://cdn.example.com/bootstrap.min.css" integrity="sha384-xyz" crossorigin="anonymous">
```

9. Real-World Case Study / Example

- **Case Study:** Attaches external CSS stylesheets across all production web applications.

```
<link rel="stylesheet" href="print.css" media="print">
```

10. Important Notes

Must be placed inside <head>. Void tag (no closing </link> tag).

11. Related Tags

Related to <style> and <script>.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Standard element.

MODULE 2: HEADINGS, TEXT FORMATTING & LINKS

2.1 <h1 to h6>

1. Tag Name

- **Tag Name:** <h1 to h6>
- **Opening & Closing Form:** <h1>...</h1> to <h6>...</h6>
- **Tag Type:** Paired / Container Tags

2. Definition

The <h1> through <h6> elements represent six levels of document section headings. <h1> is the highest (most important) level, and <h6> is the lowest level.

3. Purpose / Function

Structures document outline hierarchy, improves content readability, and signals section topics to search engines and screen readers.

4. Syntax

```
<h1>Heading Level 1</h1>
<h2>Heading Level 2</h2>
<h3>Heading Level 3</h3>
```

5. Basic Example

```
<h1>Web Technologies (18CS52)</h1>
<h2>Module 1: HTML5 Syntax</h2>
<h3>1.1 Tags Overview</h3>
```

6. Output

Renders bold heading text blocks with default font sizes decreasing from

(largest) to

(smallest).

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
align	Sets heading alignment (Obsolete).	left, center, right, justify	Obsolete (Use CSS)
Global Attributes	Standard global attributes (id, class, style, title, dir, lang, hidden, tabindex, accesskey, role).	Standard global values	Optional

8. Attribute Code Example

```
<h1 id="top-title" class="text-primary font-bold" dir="ltr" tabindex="0">CSE Department</h1>
```

9. Real-World Case Study / Example

- **Case Study:** Used for page article titles, section headers, and documentation headings.

```
<article>
  <h1 id="ds-lab">Data Structures Lab</h1>
  <h2>Binary Search Trees</h2>
</article>
```

10. Important Notes

Use headings in natural descending sequence (<h1> -> <h2> -> <h3>). Avoid skipping heading levels.

11. Related Tags

Related to <p>, <section>, <header> .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. align attribute (<h1 align="center">) is obsolete; use CSS text-align instead.

2.2 <p>

1. Tag Name

- **Tag Name:** <p>
- **Opening & Closing Form:** <p>...</p>
- **Tag Type:** Paired / Container Tag

2. Definition

The <p> element represents a paragraph block of text.

3. Purpose / Function

Groups text sentences logically, automatically creating vertical blank line margins before and after the text block.

4. Syntax

```
<p>Paragraph text content...</p>
```

5. Basic Example

```
<p>HTML5 is the fifth and current major version of the HTML standard.</p>
<p>It includes new semantic layout elements like header, footer, and section.</p>
```

6. Output

Renders two distinct text paragraphs separated by vertical space margins.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
align	Paragraph alignment (Obsolete).	left, center, right, justify	Obsolete (Use CSS)
Global Attributes	Standard global attributes (id, class, style, title, dir, lang, hidden, tabindex).	Standard global values	Optional

8. Attribute Code Example

```
<p id="intro-p" class="lead text-muted" dir="ltr">Updated on August 2026.</p>
```

9. Real-World Case Study / Example

- **Case Study:** Used for all prose body text across articles, blogs, and portals.

```
<section>
  <p class="description">Student USN must be verified before login.</p>
</section>
```

10. Important Notes

Browser engines automatically add vertical block margins around <p> tags.

11. Related Tags

Related to
, <div>, .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Standard element.

2.3 <br and hr>

1. Tag Name

- **Tag Name:** <br and hr>
- **Opening & Closing Form:**
 and <hr>
- **Tag Type:** Void / Self-Closing Tags

2. Definition

 inserts a single text line break. <hr> inserts a thematic horizontal rule line divider.

3. Purpose / Function

 breaks a text line without starting a new paragraph margin. <hr> creates a visual section break line.

4. Syntax

```
<p>Line 1<br>Line 2</p>
<hr>
```

5. Basic Example

```
<h1>VTU Examination Notice</h1>
<hr>
<p>Date: 01 Sept 2026<br>Time: 9:30 AM</p>
```

6. Output

Renders a title heading, followed by a solid horizontal divider line, followed by two lines of text stacked directly on top of each other.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
align (in hr)	Rule alignment (Obsolete).	left , center , right	Obsolete (Use CSS)
width (in hr)	Rule width (Obsolete).	Pixels / percentage	Obsolete (Use CSS)
size (in hr)	Rule height/thickness (Obsolete).	Pixels	Obsolete (Use CSS)
noshade (in hr)	Solid color without shading (Obsolete).	noshade	Obsolete (Use CSS)
color (in hr)	Rule color (Obsolete).	Color hex/name	Obsolete (Use CSS)
Global Attributes	Standard global attributes.	Standard global values	Optional

8. Attribute Code Example

```
<hr id="section-divider" class="my-4 border-top">
```

9. Real-World Case Study / Example

- **Case Study:** Used for address lines, poem line breaks, and dividing article sections.

```
<address>CSE Block<br>Belagavi</address>
<hr>
```

10. Important Notes

Use
 ONLY for line breaks that are part of text content (like addresses/poetry), NOT for creating layout spacing (use CSS margins instead).

11. Related Tags

Related to <p> , <div> .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Legacy <hr width="50%" align="center"> styling attributes are obsolete in HTML5.

2.4 <a>

1. Tag Name

- **Tag Name:** <a>
- **Opening & Closing Form:** ...
- **Tag Type:** Paired Tag

2. Definition

The <a> (Anchor) element creates a hyperlink to web pages, files, email addresses, location anchors, or external URLs.

3. Purpose / Function

Forms the primary building block of the Web, enabling navigation between documents.

4. Syntax

```
<a href="URL" target="_blank">Link Text</a>
```

5. Basic Example

```
<a href="https://vtu.ac.in" target="_blank" rel="noopener noreferrer" title="VTU Portal">Visit VTU Official Website</a>
```

6. Output

Renders underlined blue text "Visit VTU Official Website". Clicking it opens vtu.ac.in in a new browser tab.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
href	Target URL or page anchor fragment link destination.	URL string (https://... , #anchor , mailto:... , tel:...)	Mandatory for links
target	Target browsing context for opening link.	_blank (new tab), _self (same tab), _parent , _top	Optional
download	Prompts browser to download target file instead of navigating.	Filename string or empty	Optional
rel	Specifies relationship between current document and target document.	noopener , noreferrer , author , bookmark , external , help , license , next , nofollow , prev , search	Security Best Practice
type	MIME type of target resource.	application/pdf , text/html	Optional
hreflang	Natural language of linked target page.	Language code (en , hi)	Optional
referrerpolicy	Controls referrer header sent with request.	no-referrer , origin , strict-origin-when-cross-origin	Optional
ping	List of URLs to notify via POST when user follows link.	Space-separated URLs	Optional
name / charset / coords / shape / rev	Legacy attributes.	Strings	Obsolete in HTML5

8. Attribute Code Example

```
<a href="syllabus.pdf" download="CSE_Syllabus.pdf" target="_blank" rel="noopener" class="btn btn-primary">Download Syllabus</a>
```

9. Real-World Case Study / Example

- **Case Study:** Used for menu links, button actions, document downloads, and internal page jump anchors (#section1).

```
<nav><a href="/home">Home</a> | <a href="/contact">Contact</a></nav>
```

10. Important Notes

Always use rel="noopener noreferrer" when setting target="_blank" to prevent security tabnabbing vulnerabilities.

11. Related Tags

Related to <nav> , <link> , .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Core hyperlink element.

2.5

1. Tag Name

- **Tag Name:**
- **Opening & Closing Form:**
- **Tag Type:** Void / Self-Closing Tag

2. Definition

The element embeds an image into an HTML document.

3. Purpose / Function

Displays graphic files (JPEG, PNG, SVG, WebP, GIF) on web pages.

4. Syntax

```

```

5. Basic Example

```

```

6. Output

Renders a 120x120 pixel image of the VTU logo on the webpage layout.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
src	Path location URL of image file resource.	Relative or absolute URL	Mandatory
alt	Accessible fallback text description for screen readers or broken links.	Descriptive text string	Mandatory (for Accessibility)
width	Display width in pixels.	Integer pixels	Recommended
height	Display height in pixels.	Integer pixels	Recommended
loading	Configures browser image loading performance strategy.	lazy (loads when near viewport), eager (loads immediately)	Optional
srcset	List of image sources for responsive pixel densities.	Candidate image URLs with width/density descriptors	Optional
sizes	Image layout sizes for responsive srcset evaluation.	Media condition size strings	Optional with srcset
decoding	Hints browser on image decoding mode.	async, sync, auto	Optional
fetchpriority	Sets download priority hint for browser preloader.	high, low, auto	Optional
usemap	Associates image with an interactive <map> element.	#mapname	Optional
ismap	Specifies image as a server-side image map.	ismap	Optional
crossorigin	Configures CORS request settings.	anonymous, use-credentials	Optional
referrerpolicy	Specifies referrer policy when fetching image.	no-referrer, strict-origin	Optional
align / border / hspace / vspace / name	Presentation attributes.	Various	Obsolete in HTML5

8. Attribute Code Example

```

```

9. Real-World Case Study / Example

- **Case Study:** Used for website logos, user profile avatars, product photos, and diagram figures.

```
<figure>
  
</figure>
```

10. Important Notes

Always provide a descriptive alt attribute. If image is purely decorative, use alt="" .

11. Related Tags

Related to <picture> , <figure> , <figcaption> .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Core media tag.

2.6 <strong and em>

1. Tag Name

- **Tag Name:** <strong and em>
- **Opening & Closing Form:** ... and ...
- **Tag Type:** Paired Inline Tags

2. Definition

 indicates strong semantic importance. indicates vocal stress emphasis.

3. Purpose / Function

 renders text in bold to signal critical urgency. renders text in italics to signal verbal emphasis.

4. Syntax

```
<strong>Important Text</strong>
<em>Emphasized Text</em>
```

5. Basic Example

```
<p><strong>WARNING:</strong> Submission deadline is <em>today</em> before 12:00 PM.</p>
```

6. Output

Renders "WARNING:" in bold and "today" in italics. Screen readers pronounce with elevated emphasis.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
Global Attributes	Standard global attributes (id, class, style, title, dir, lang, hidden).	Standard global values	Optional

8. Attribute Code Example

```
<strong id="warn-msg" class="text-danger" title="Important Notice">Exam Hall Ticket Required!</strong>
```

9. Real-World Case Study / Example

- **Case Study:** Used in documentation warnings, key search terms, and emphasis in text body.

```
<p><strong>Note:</strong> Always save your file before compiling.</p>
```

10. Important Notes

`<strong>` vs `<b>` : `<b>` is purely visual bold; `<strong>` means semantic importance. `<em>` vs `<i>` : `<i>` is visual italic; `<em>` is vocal stress.

11. Related Tags

Related to `<b>` , `<i>` , `<mark>` .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Preferred semantic tags.

MODULE 3: LIST TAGS

3.1 `<ul>` and `<ol>`

1. Tag Name

- **Tag Name:** `<ul>` and `<ol>`
- **Opening & Closing Form:** `<ul>...</ul>` and `<ol>...</ol>`
- **Tag Type:** Paired Container Tags

2. Definition

`<ul>` defines an unordered bulleted list. `<ol>` defines an ordered numbered list. Each item inside is enclosed in `<li>` (List Item).

3. Purpose / Function

Groups related items sequentially (ordered sequence) or un-ordered (bullet points).

4. Syntax

```
<ul>
  <li>Item</li>
</ul>
<ol>
  <li>Item</li>
</ol>
```

5. Basic Example

```
<h3>Required Tools:</h3>
<ul>
  <li>VS Code IDE</li>
  <li>Node.js Engine</li>
</ul>
<h3>Lab Execution Steps:</h3>
<ol type="1" start="1">
  <li>Write HTML Code</li>
  <li>Open in Chrome Browser</li>
</ol>
```

6. Output

Renders a bulleted list of tools and a numbered list (1, 2) of execution steps.

7. Exhaustive Attributes Reference Table

Tag	Attribute	Description	Valid Values	Required/Optional
<code></code>	<code>type</code>	Numbering type style.	<code>1</code> (numbers), <code>A</code> (uppercase alpha), <code>a</code> (lowercase alpha), <code>I</code> (uppercase Roman), <code>i</code> (lowercase Roman)	Optional
<code></code>	<code>start</code>	Starting integer number value for list ordering.	Integer number	Optional
<code></code>	<code>reversed</code>	Reverses list ordering sequence.	<code>reversed</code>	Optional
<code></code>	<code>value</code>	Sets explicit integer value for current item inside <code></code> .	Integer number	Optional
<code></code>	<code>type</code>	Bullet marker shape (Obsolete).	<code>disc</code> , <code>circle</code> , <code>square</code>	Obsolete (Use CSS)
<code></code>	<code>compact</code>	Compact display layout (Obsolete).	<code>compact</code>	Obsolete (Use CSS)

8. Attribute Code Example

```
<ol type="I" start="1" reversed class="syllabus-list">
  <li value="5">Step 5</li>
</ol>
```

9. Real-World Case Study / Example

- **Case Study:** Used for navigation menus (`<ul>`), recipe steps (`<ol>`), and dropdown lists.

```
<nav>
  <ul>
    <li><a href="#home">Home</a></li>
    <li><a href="#about">About</a></li>
  </ul>
</nav>
```

10. Important Notes

Direct children of `<ul>` and `<ol>` MUST ONLY be `<li>` elements.

11. Related Tags

Contains `<li>` . Related to `<dl>` .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. `type` attribute on `<ul>` (`disc` , `circle`) is deprecated in HTML5; use CSS `list-style-type` instead.

3.2 `<dl, dt, dd>`

1. Tag Name

- **Tag Name:** `<dl, dt, dd>`
- **Opening & Closing Form:** `<dl><dt>...</dt><dd>...</dd></dl>`
- **Tag Type:** Paired Container Tags

2. Definition

The `<dl>` tag defines a description list. `<dt>` specifies the term/name, and `<dd>` specifies the description/value of that term.

3. Purpose / Function

Used to display glossaries, dictionary definitions, or key-value metadata pairs. *(Frequently asked in 10-mark exam questions).*

4. Syntax

```
<dl>
  <dt>Term 1</dt>
  <dd>Description 1</dd>
</dl>
```

5. Basic Example

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language - Standard language for web documents.</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets - Language for document presentation styling.</dd>
</dl>
```

6. Output

Renders terms "HTML" and "CSS" in bold text, followed by indented description lines beneath each term.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
Global Attributes	Standard global attributes (<code>id</code> , <code>class</code> , <code>style</code> , <code>dir</code> , <code>lang</code>).	Standard global values	Optional

8. Attribute Code Example

```
<dl id="tech-glossary" class="row">
  <dt class="col-sm-3">Course Code</dt>
  <dd class="col-sm-9">18CS52</dd>
</dl>
```

9. Real-World Case Study / Example

- **Case Study:** Used for FAQs, glossary definitions, metadata terms, and key-value specs.

```
<dl>
  <dt>Author</dt><dd>Tim Berners-Lee</dd>
  <dt>Year</dt><dd>1991</dd>
</dl>
```

10. Important Notes

Each `<dt>` can have one or multiple matching `<dd>` description tags.

11. Related Tags

Related to `<ul>` , `<ol>` .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Core semantic list structure.

MODULE 4: TABLES (LAB VIVA & EXAMINATION CORE)

4.1 <table>, <tr>, <th>, <td>, <caption>

1. Tag Name

- **Tag Name:** <table>, <tr>, <th>, <td>, <caption>
- **Opening & Closing Form:** <table>...</table>
- **Tag Type:** Paired Container Tags

2. Definition

The <table> element represents data in a two-dimensional grid of rows (<tr>) and cells (<th> header cell, <td> data cell), with an optional title caption (<caption>).

3. Purpose / Function

Presents structured tabular data (schedules, grade sheets, database rows).

4. Syntax

```
<table>
  <caption>Title</caption>
  <tr>
    <th>Header</th>
  </tr>
  <tr>
    <td>Data</td>
  </tr>
</table>
```

5. Basic Example

```
<table border="1">
  <caption>CSE Semester 3 Timetable</caption>
  <tr>
    <th>USN</th>
    <th>Name</th>
    <th colspan="2">Marks (Theory + Lab)</th>
  </tr>
  <tr>
    <td>1VU22CS001</td>
    <td>Aryansh</td>
    <td>48</td>
    <td>98</td>
  </tr>
</table>
```

6. Output

Renders a 2-row table grid with borderlines, bold centered column headers, and merged header cell spanning across two columns.

7. Exhaustive Attributes Reference Table

Tag	Attribute	Description	Valid Values	Required/Optional
<th> / <td>	colspan	Merges <i>n</i> adjacent columns horizontally.	Positive integer number (1 , 2 , 3)	Highly Tested in Exams
<th> / <td>	rowspan	Merges <i>n</i> adjacent rows vertically.	Positive integer number (1 , 2 , 3)	Highly Tested in Exams
<th>	scope	Associates header cell with columns or rows.	col , row , colgroup , rowgroup	Accessibility Best Practice
<th> / <td>	headers	Associates cell with explicit header ID string.	Header ID string	Optional
<th>	abbr	Short abbreviation text for screen readers.	Abbreviation string	Optional
<table>	border	Table border width (Only 1 or empty in HTML5).	1 or ""	Optional (Use CSS)
<table>	cellpadding	Cell internal padding (Obsolete).	Integer pixels	Obsolete (Use CSS)
<table>	cellspacing	Cell border spacing (Obsolete).	Integer pixels	Obsolete (Use CSS)
<table>	width / height	Table dimensions (Obsolete).	Pixels / percentage	Obsolete (Use CSS)
<table> / <tr> / <th> / <td>	align / valign	Alignment (Obsolete).	left , center , right , top , middle , bottom	Obsolete (Use CSS)
<table>	bgcolor	Background color (Obsolete).	Color hex/name	Obsolete (Use CSS)

8. Attribute Code Example

```
<td rowspan="2" colspan="2" headers="th-marks" class="bg-warning">Merged Cell</td>
```

9. Real-World Case Study / Example

- **Case Study:** Used for grade reports, examination timetables, product pricing tables, and financial data grids.

```
<table>
  <thead><tr><th scope="col">ID</th><th scope="col">Name</th></tr></thead>
  <tbody><tr><td>1</td><td>Alice</td></tr></tbody>
</table>
```

10. Important Notes

Do NOT use tables for page layout! Use CSS Flexbox/Grid for layout, and <table> ONLY for tabular data.

11. Related Tags

Contains `<caption>`, `<thead>`, `<tbody>`, `<tfoot>`, `<tr>`, `<th>`, `<td>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Layout presentation attributes (`cellpadding`, `cellspacing`, `width`, `bgcolor`, `align`) are obsolete in HTML5; use CSS styling instead.

MODULE 5: FORMS AND INPUT CONTROLS (HIGHEST EXAM WEIGHTAGE)

5.1 `<form>`

1. Tag Name

- **Tag Name:** `<form>`
- **Opening & Closing Form:** `<form>...</form>`
- **Tag Type:** Paired Container Tag

2. Definition

The `<form>` element defines an interactive HTML form used to collect user inputs and send the data payload to a processing web server.

3. Purpose / Function

Acts as the primary wrapper control holding input fields, textareas, checkboxes, dropdowns, and submit buttons.

4. Syntax

```
<form action="server-endpoint" method="POST">
  <!-- Form controls -->
</form>
```

5. Basic Example

```
<form action="/submit-student.php" method="POST" enctype="multipart/form-data" autocomplete="on">
  <label for="usn">USN:</label>
  <input type="text" id="usn" name="student_usn" required>
  <button type="submit">Submit</button>
</form>
```

6. Output

Renders an interactive form containing a text input field and a clickable submit button.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>action</code>	URL of server script endpoint that processes form data.	URL string (<code>/submit.php</code> , <code>https://...</code>)	Mandatory for submission
<code>method</code>	HTTP submission method.	<code>GET</code> (appends in URL query), <code>POST</code> (sends in HTTP body payload), <code>dialog</code>	Mandatory
<code>enctype</code>	Encoding type for form submission payload.	<code>application/x-www-form-urlencoded</code> (default), <code>multipart/form-data</code> (for file uploads), <code>text/plain</code>	Required for File Uploads
<code>target</code>	Target browsing context window for receiving submission response.	<code>_self</code> , <code>_blank</code> , <code>_parent</code> , <code>_top</code>	Optional
<code>autocomplete</code>	Toggles browser auto-fill for all child inputs.	<code>on</code> , <code>off</code>	Optional
<code>novalidate</code>	Disables native HTML5 browser form validation checks.	<code>novalidate</code>	Optional
<code>name</code>	Unique name identifier for form.	Name string	Optional
<code>rel</code>	Relationship of form submit link target.	<code>external</code> , <code>help</code> , <code>license</code>	Optional
<code>accept-charset</code>	Character encodings accepted by server.	<code>UTF-8</code>	Optional

8. Attribute Code Example

```
<form action="/upload" method="POST" enctype="multipart/form-data" autocomplete="on" novalidate class="card p-4">
```

9. Real-World Case Study / Example

- **Case Study:** Used for user registration forms, login screens, search bars, checkout forms, and surveys.

```
<form action="/api/v1/register" method="POST">
  <input type="email" name="email">
  <input type="submit" value="Register">
</form>
```

10. Important Notes

Difference between `GET` and `POST` is a classic 5-mark exam question! `GET` is bookmarkable and exposes data in URL bar; `POST` is secure and handles large payload/file uploads.

11. Related Tags

Contains `<input>`, `<label>`, `<button>`, `<select>`, `<textarea>`, `<fieldset>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Core web form container.

5.2 <input>

1. Tag Name

- **Tag Name:** <input>
- **Opening & Closing Form:** <input>
- **Tag Type:** Void / Self-Closing Tag

2. Definition

The <input> element is the most vital interactive form control, allowing users to enter data in various formats specified by the type attribute.

3. Purpose / Function

Renders text boxes, password fields, radio buttons, checkboxes, file upload pickers, date pickers, range sliders, and submit buttons.

4. Syntax

```
<input type="type_name" name="variable_name" value="val">
```

5. Basic Example

```
<p>USN: <input type="text" name="usn" placeholder="1VU22CS001" required></p>
<p>Password: <input type="password" name="pass" required></p>
<p>Gender: <input type="radio" name="g" value="m" checked> M <input type="radio" name="g" value="f"> F</p>
<p><input type="submit" value="Register"></p>
```

6. Output

Renders a text box with placeholder, a masked password field, two radio buttons (male pre-checked), and a submit button.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>type</code>	Type of interactive input control to render.	<code>text</code> , <code>password</code> , <code>email</code> , <code>number</code> , <code>checkbox</code> , <code>radio</code> , <code>file</code> , <code>hidden</code> , <code>submit</code> , <code>reset</code> , <code>button</code> , <code>date</code> , <code>time</code> , <code>datetime-local</code> , <code>month</code> , <code>week</code> , <code>color</code> , <code>range</code> , <code>search</code> , <code>tel</code> , <code>url</code> , <code>image</code>	Mandatory
<code>name</code>	Field variable key name submitted in HTTP payload.	Variable name string	Mandatory for Server Submission
<code>value</code>	Initial or submitted data value string.	Value string	Required for radio/checkbox/submit
<code>placeholder</code>	Short hint text displayed inside empty text box.	Hint string	Recommended
<code>required</code>	Mandates field completion before submission.	<code>required</code>	Mandatory for required inputs
<code>readonly</code>	Prevents user modification while remaining selectable.	<code>readonly</code>	Optional
<code>disabled</code>	Deactivates control and excludes it from payload.	<code>disabled</code>	Optional
<code>checked</code>	Pre-checks radio button or checkbox control.	<code>checked</code>	Optional (radio/checkbox)
<code>pattern</code>	Regular Expression (Regex) string for format validation.	Regex pattern string (<code>[A-Z]{3}[0-9]{3}</code>)	Optional
<code>minlength</code> / <code>maxlength</code>	Specifies character length boundaries.	Positive integer number	Optional
<code>min</code> / <code>max</code> / <code>step</code>	Specifies numeric or date range boundaries and step increment.	Number or date string	Optional (number/range/date)
<code>accept</code>	Specifies accepted file extensions/MIME types.	<code>.pdf</code> , <code>.zip</code> , <code>image/*</code>	Optional (file type)
<code>multiple</code>	Allows selecting multiple files or email addresses.	<code>multiple</code>	Optional (file/email)
<code>autocomplete</code>	Enables browser auto-fill for this specific input.	<code>on</code> , <code>off</code> , <code>username</code> , <code>current-password</code>	Optional
<code>autofocus</code>	Automatically focuses input on page load.	<code>autofocus</code>	Optional
<code>list</code>	Associates input with a <code><datalist></code> element ID.	Datalist ID string	Optional
<code>form</code>	Binds input to an external form ID outside parent form.	Form ID string	Optional
<code>formaction</code> / <code>formmethod</code> / <code>formenctype</code> / <code>formnovalidate</code> / <code>formtarget</code>	Overrides parent form submission attributes for submit buttons.	Various	Optional (submit buttons)
<code>src</code> / <code>alt</code> / <code>width</code> / <code>height</code>	Image properties for <code>type="image"</code> .	Image paths/dimensions	Optional (<code>type="image"</code>)
<code>capture</code>	Specifies media capture camera/mic for mobile upload.	<code>user</code> , <code>environment</code>	Optional (<code>type="file"</code>)

8. Attribute Code Example

```
<input type="email" name="user_email" id="user_email" class="form-control" placeholder="user@college.edu" required autocomplete="email" autofocus>
```

9. Real-World Case Study / Example

- **Case Study:** Used across all forms on the internet for data input collection.

```
<input type="file" name="assignment" accept=".pdf,.zip" multiple required>
```

10. Important Notes

Input controls without a `name` attribute will NOT be submitted to the server during form POST/GET!

11. Related Tags

Paired with `<label for="...">`. Container is `<form>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. HTML5 added new input types (`email`, `url`, `number`, `range`, `date`, `color`, `search`).

5.3 `<label>`

1. Tag Name

- **Tag Name:** `<label>`
- **Opening & Closing Form:** `<label for="id">...</label>`

- **Tag Type:** Paired Tag

2. Definition

The `<label>` element represents a caption for an item in a user interface.

3. Purpose / Function

Improves web accessibility for screen readers and increases clickable hit target area for radio buttons and checkboxes.

4. Syntax

```
<label for="target_input_id">Label Text</label>
<input type="text" id="target_input_id" name="var">
```

5. Basic Example

```
<label for="usn_field">Student USN:</label>
<input type="text" id="usn_field" name="usn">
```

6. Output

Renders text "Student USN:". Clicking the text "Student USN:" automatically focuses the text cursor inside the input field.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>for</code>	Specifies the <code>id</code> of the target input element this label belongs to.	Target Input ID string	Mandatory
<code>form</code>	Associates label with an external form ID.	Form ID string	Optional

8. Attribute Code Example

```
<label for="agree_check" class="form-check-label">I agree to terms</label>
```

9. Real-World Case Study / Example

- **Case Study:** Mandatory web accessibility standard on every form input.

```
<label for="pass">Password:</label>
<input type="password" id="pass" name="password">
```

10. Important Notes

The `for` attribute value MUST match the target input's `id` attribute value exactly!

11. Related Tags

Related to `<input>`, `<select>`, `<textarea>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Accessibility core tag.

5.4 `<select and option>`

1. Tag Name

- **Tag Name:** `<select and option>`
- **Opening & Closing Form:** `<select><option>...</option></select>`
- **Tag Type:** Paired Container Tags

2. Definition

The `<select>` element defines a drop-down menu list. The `<option>` elements define the selectable list items.

3. Purpose / Function

Provides a drop-down menu for users to select one (or multiple) choices from a defined set of options.

4. Syntax

```
<select name="var">
  <option value="val1">Label 1</option>
</select>
```

5. Basic Example

```
<label for="branch">Select Branch:</label>
<select id="branch" name="student_branch" required>
  <option value="" disabled selected>-- Choose Branch --</option>
  <option value="cse">Computer Science (CSE)</option>
  <option value="ise">Information Science (ISE)</option>
  <option value="ece">Electronics (ECE)</option>
</select>
```

6. Output

Renders a drop-down menu displaying selectable engineering branches.

7. Exhaustive Attributes Reference Table

Tag	Attribute	Description	Valid Values	Required/Optional
<select>	name	Field variable key sent in HTTP payload.	Variable name string	Mandatory
<select>	multiple	Allows selecting multiple options simultaneously.	multiple	Optional
<select>	size	Specifies visible height count of dropdown list lines.	Positive integer	Optional
<select>	required	Mandates option selection before form submission.	required	Optional
<select>	disabled	Deactivates dropdown selection control.	disabled	Optional
<select>	autofocus	Focuses dropdown automatically on page load.	autofocus	Optional
<select>	form	Associates select box with an external form ID.	Form ID string	Optional
<option>	value	Data payload value string submitted when selected.	Value string	Mandatory
<option>	selected	Sets option as default pre-selected item on load.	selected	Optional
<option>	disabled	Deactivates option from selection.	disabled	Optional
<option>	label	Short label text for option element.	Label string	Optional

8. Attribute Code Example

```
<select name="course" id="course" class="form-select" required size="1">
  <option value="3" selected>Semester 3</option>
</select>
```

9. Real-World Case Study / Example

- **Case Study:** Used for country pickers, branch selectors, payment method dropdowns, and state lists.

```
<select name="sem">
  <option value="3" selected>Semester 3</option>
</select>
```

10. Important Notes

If an <option> has no value attribute specified, the text inside option is sent to server as value.

11. Related Tags

Parent <select>, child <option>, <optgroup>.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Standard dropdown control.

5.5 <textarea>

1. Tag Name

- **Tag Name:** <textarea>
- **Opening & Closing Form:** <textarea>...</textarea>
- **Tag Type:** Paired Tag

2. Definition

The <textarea> element represents a multi-line plain-text editing control.

3. Purpose / Function

Allows users to enter large multi-line text input (comments, addresses, feedback, long answers).

4. Syntax

```
<textarea name="var" rows="4" cols="50">Default text</textarea>
```

5. Basic Example

```
<label for="feedback">Student Remarks:</label><br>
<textarea id="feedback" name="remarks" rows="4" cols="40" placeholder="Enter comments here..."></textarea>
```

6. Output

Renders a 4-row by 40-column resizable multi-line text box with placeholder text.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
name	Field variable key for server submission payload.	Variable name string	Mandatory
rows	Visible height of text area in text lines count.	Positive integer number	Recommended
cols	Visible width of text area in average character widths.	Positive integer number	Recommended
placeholder	Hint text shown inside empty text box.	Hint string	Optional
required	Mandates field completion before form submission.	required	Optional
readonly	Prevents text editing while remaining selectable.	readonly	Optional
disabled	Deactivates text area control.	disabled	Optional
minlength / maxlength	Specifies character length boundaries.	Positive integer number	Optional
wrap	Controls text wrapping on submission (soft default, hard inserts newlines).	soft , hard	Optional
autofocus	Focuses text area on page load.	autofocus	Optional
form	Associates text area with an external form ID.	Form ID string	Optional

8. Attribute Code Example

```
<textarea name="bio" id="bio" class="form-control" rows="3" cols="50" placeholder="Short Bio" maxlength="500" required></textarea>
```

9. Real-World Case Study / Example

- **Case Study:** Used for comment sections, feedback forms, essay submissions, and address fields.

```
<textarea name="code_submission" rows="10" cols="80"></textarea>
```

10. Important Notes

Default initial text must be placed BETWEEN `<textarea>` and `</textarea>` , NOT inside a `value` attribute!

11. Related Tags

Related to `<input>` , `<form>` .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Standard multi-line input tag.

MODULE 6: SEMANTIC LAYOUT TAGS (HTML5 CORE)

6.1 `<header, footer, main, section, article, aside, nav>`

1. Tag Name

- **Tag Name:** `<header, footer, main, section, article, aside, nav>`
- **Opening & Closing Form:** `<semantic-tag>...</semantic-tag>`
- **Tag Type:** Paired Container Tags

2. Definition

HTML5 semantic layout elements give clear meaning to web page structures, replacing generic non-semantic `<div>` tags with meaningful layout landmarks.

3. Purpose / Function

Improves website SEO, page structure organization, readability, and accessibility for screen reader assistive software.

4. Syntax

```
<header>Nav & Title</header>
<main>
  <section>
    <article>Content</article>
  </section>
  <aside>Sidebar</aside>
</main>
<footer>Copyright</footer>
```

5. Basic Example

```
<header>
  <h1>VTU CSE Department</h1>
  <nav aria-label="Main Menu"><a href="#home">Home</a></nav>
</header>
<main id="main-content" role="main">
  <section id="assignment">
    <h2>Web Tech Assignment</h2>
    <p>Complete all module questions.</p>
  </section>
</main>
<footer>
  <p>&copy; 2026 VTU. All rights reserved.</p>
</footer>
```

6. Output

Renders structured page layout with header banner at top, main section in middle, and footer banner at bottom.

7. Exhaustive Attributes Reference Table

Tag Name	Semantic Purpose & Function	Common Use Case
<header>	Introductory branding, site logo, title & navigation.	Top bar header of website or article header.
<nav>	Major section containing primary navigation links.	Top navbar, menu list links.
<main>	Unique primary landmark content area of the document.	Main body content (Only 1 per page).
<section>	Standalone thematic section with a heading.	Landing page sections (Features, Pricing, Contact).
<article>	Self-contained, independently distributable content unit.	Blog posts, news articles, forum comments.
<aside>	Content indirectly related to main content.	Blog sidebars, callout boxes, advertisements.
<footer>	Footer for page or section (copyright, contact).	Bottom site footer, copyright notice.
Global Attributes	Standard global attributes (id , class , style , role , aria-label , hidden).	Standard global values

8. Attribute Code Example

```
<main id="main-content" role="main" aria-label="Primary Content" class="container py-4">
```

9. Real-World Case Study / Example

- **Case Study:** Used on 100% of modern HTML5 websites to build clean semantic document layouts.

```
<header><nav>Links</nav></header>
<main><article>Post</article></main>
<footer>Copyright</footer>
```

10. Important Notes

<div> vs Semantic Tags: <div> has ZERO semantic meaning (used only for CSS styling). <section> and <article> tell search engines what the content actually represents.

11. Related Tags

Related to <div> , .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Native HTML5 landmark elements.

6.2 <div and span>

1. Tag Name

- **Tag Name:** <div and span>
- **Opening & Closing Form:** <div>...</div> and ...
- **Tag Type:** Paired Container Tags

2. Definition

<div> (Content Division) is a generic **block-level** container. is a generic **inline** container. Neither carries semantic meaning.

3. Purpose / Function

Used to group elements or text fragments for CSS styling (colors, layout) or JavaScript DOM manipulation when no semantic tag applies.

4. Syntax

```
<div class="box">
  <p>Text with <span class="highlight">colored</span> word.</p>
</div>
```

5. Basic Example

```
<div style="background-color: #f1f5f9; padding: 15px; border-radius: 5px;">
  <h3>Notification</h3>
  <p>Status: <span style="color: green; font-weight: bold;">PASSED</span></p>
</div>
```

6. Output

Renders a full-width grey background box containing heading and text with the word "PASSED" highlighted in bold green font.

7. Exhaustive Attributes Reference Table

Feature	<div> Tag	 Tag
Display Mode	display: block;	display: inline;
Line Break	Forces new line before and after.	Fits inline inside surrounding text.
Width	Takes 100% full container width.	Occupies ONLY as much width as text needs.
Use Case	Layout wrappers, card boxes, grid rows.	Highlighting text words, badge labels, icons.
Global Attributes	Full support (id , class , style , title , dir , lang , hidden , tabindex , contenteditable , draggable , data-* , role , aria-*).	Standard global values

8. Attribute Code Example

```
<div id="card-1" class="row" data-card-id="101" role="region" tabindex="0">
  <span class="badge bg-primary" data-status="new">New</span>
</div>
```

9. Real-World Case Study / Example

- **Case Study:** Used extensively in CSS Flexbox/Grid layouts, card components, and inline text formatting.

```
<div class="card">
  <p>Price: <span class="price">$99</span></p>
</div>
```

10. Important Notes

Use semantic tags (`<article>` , `<section>` , `<header>`) whenever possible, and reserve `<div>` / `<span>` strictly for CSS styling and JS hooks.

11. Related Tags

Related to `<section>` , `<article>` , `<p>` .

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Generic structural wrappers.

MODULE 7: MULTIMEDIA, INTERACTIVE & SCRIPTING TAGS

7.1 `<audio and video>`

1. Tag Name

- **Tag Name:** `<audio and video>`
- **Opening & Closing Form:** `<video>...</video>` and `<audio>...</audio>`
- **Tag Type:** Paired Container Tags

2. Definition

The `<video>` and `<audio>` tags embed native video clips and audio streams into an HTML document without requiring third-party plugins (like Flash).

3. Purpose / Function

Plays video files (MP4, WebM) and audio files (MP3, WAV) natively in browser pages.

4. Syntax

```
<video src="file.mp4" controls width="400"></video>
<audio src="file.mp3" controls></audio>
```

5. Basic Example

```
<video width="320" height="240" controls poster="thumbnail.jpg" preload="metadata">
  <source src="lecture.mp4" type="video/mp4">
  <source src="lecture.webm" type="video/webm">
  Your browser does not support HTML5 video.
</video>
```

6. Output

Renders a video player box with media playback controls (Play, Pause, Volume, Fullscreen) displaying thumbnail image.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>src</code>	Direct path URL of media file.	URL string	Optional (if <code><source></code> used)
<code>controls</code>	Displays browser native playback controls.	<code>controls</code>	Recommended
<code>autoplay</code>	Plays media automatically on page load.	<code>autoplay</code>	Optional
<code>loop</code>	Automatically restarts media playback when finished.	<code>loop</code>	Optional
<code>muted</code>	Mutes audio output by default.	<code>muted</code>	Optional
<code>preload</code>	Hints browser on media preloading strategy.	<code>auto</code> (preloads full media), <code>metadata</code> (preloads duration/dimensions), <code>none</code>	Recommended
<code>poster</code> (in video)	Thumbnail image URL displayed before play.	Image URL string	Optional
<code>width</code> / <code>height</code> (in video)	Display dimensions in pixels.	Integer pixels	Recommended
<code>playsinline</code> (in video)	Plays inline on mobile iOS without fullscreen.	<code>playsinline</code>	Optional
<code>crossorigin</code>	Configures CORS request settings.	<code>anonymous</code> , <code>use-credentials</code>	Optional

8. Attribute Code Example

```
<video controls autoplay muted loop preload="metadata" poster="preview.jpg" width="100%" height="auto" playsinline>
```

9. Real-World Case Study / Example

- **Case Study:** Used on video streaming portals (YouTube, Netflix), podcast hosts, and course portals.

```
<audio controls preload="none">
  <source src="podcast.mp3" type="audio/mpeg">
</audio>
```

10. Important Notes

Always include `<source>` tags with multiple video formats (MP4, WebM) inside `<video>` for cross-browser fallback support.

11. Related Tags

Contains `<source>`, `<track>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Native media elements.

7.2 `<iframe>`

1. Tag Name

- **Tag Name:** `<iframe>`
- **Opening & Closing Form:** `<iframe></iframe>`
- **Tag Type:** Paired Tag

2. Definition

The `<iframe>` (Inline Frame) element is used to embed another HTML document or external webpage inside the current document.

3. Purpose / Function

Embeds external content such as Google Maps location frames, YouTube video embeds, PDF previews, or payment gateways.

4. Syntax

```
<iframe src="URL" width="width" height="height" title="desc"></iframe>
```

5. Basic Example

```
<iframe src="https://www.youtube.com/embed/dQw4w9WgXcQ" width="560" height="315" title="YouTube video player" allowfullscreen loading="lazy"></iframe>
```

6. Output

Renders an embedded 560x315 interactive YouTube video window directly on the page.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>src</code>	URL location of external webpage or document to embed.	URL string	Mandatory
<code>width</code> / <code>height</code>	Display dimensions of iframe window in pixels or %.	Integer pixels / percentage	Recommended
<code>title</code>	Accessible title description of iframe content.	Title string	Mandatory for Accessibility
<code>name</code>	Target browsing context window name for target links.	Name string	Optional
<code>sandbox</code>	Applies security restrictions to embedded frame.	<code>allow-forms</code> , <code>allow-scripts</code> , <code>allow-same-origin</code> , <code>allow-popups</code> , <code>allow-modals</code> , <code>allow-top-navigation</code>	Security Best Practice
<code>loading</code>	Configures lazy loading performance.	<code>lazy</code> , <code>eager</code>	Optional
<code>allow</code>	Specifies Feature Policy / Permissions Policy permissions.	<code>camera</code> , <code>microphone</code> , <code>geolocation</code> , <code>fullscreen</code> , <code>autoplay</code>	Optional
<code>allowfullscreen</code>	Permits iframe to request fullscreen display.	<code>allowfullscreen</code>	Optional
<code>referrerpolicy</code>	Controls referrer header sent when loading frame.	<code>no-referrer</code> , <code>strict-origin</code>	Optional
<code>frameborder</code> / <code>marginwidth</code> / <code>marginheight</code> / <code>scrolling</code> / <code>align</code>	Presentation attributes.	Various	Obsolete in HTML5

8. Attribute Code Example

```
<iframe src="map.html" width="100%" height="450" title="Campus Map" loading="lazy" sandbox="allow-scripts allow-same-origin" allowfullscreen>
```

9. Real-World Case Study / Example

- **Case Study:** Used for YouTube video embeds, Google Maps widgets, and embedded payment frames (Stripe, Razorpay).

```
<iframe src="document.pdf" width="600" height="400" title="PDF Reader"></iframe>
```

10. Important Notes

Always add a `title` attribute to `<iframe>` tags so screen readers can describe the embedded content to visually impaired users.

11. Related Tags

Related to `<video>`, `<a>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Standard embedding frame tag.

7.3 `<script and noscript>`

1. Tag Name

- **Tag Name:** `<script and noscript>`
- **Opening & Closing Form:** `<script>...</script>`
- **Tag Type:** Paired Tags

2. Definition

`<script>` embeds dynamic executable JavaScript code or links external script files. `<noscript>` defines fallback content rendered when JavaScript is disabled.

3. Purpose / Function

`<script>` adds dynamic interactivity, form validation, and DOM manipulation. `<noscript>` warns users if JS is turned off.

4. Syntax

```
<script src="app.js" defer></script>
<noscript>Please enable JavaScript.</noscript>
```

5. Basic Example

```
<script>
  function showMessage() {
    alert("Assignment Submitted!");
  }
</script>
<noscript>
  <p style="color:red;">JavaScript is required to run this application.</p>
</noscript>
```

6. Output

Executes JavaScript logic invisibly. Renders warning text ONLY if browser has disabled JavaScript execution.

7. Exhaustive Attributes Reference Table

Attribute	Description	Valid Values	Required/Optional
<code>src</code>	Path to external JavaScript file (.js).	URL string	Mandatory for external scripts
<code>defer</code>	Downloads script asynchronously and executes ONLY after HTML parsing finishes.	<code>defer</code>	Recommended for Head scripts
<code>async</code>	Downloads script asynchronously and executes immediately when loaded.	<code>async</code>	Optional
<code>type</code>	MIME type or script module context.	<code>text/javascript</code> , <code>module</code> , <code>importmap</code>	Optional
<code>crossorigin</code>	Configures CORS request behavior.	<code>anonymous</code> , <code>use-credentials</code>	Optional
<code>integrity</code>	Subresource Integrity (SRI) security verification hash string.	Hash string (<code>sha384-...</code>)	Optional
<code>nomodule</code>	Prevents script execution in browsers that support ES modules.	<code>nomodule</code>	Optional
<code>referrerpolicy</code>	Controls referrer header sent when fetching script.	<code>no-referrer</code> , <code>origin</code>	Optional
<code>language</code>	Scripting language (Obsolete).	<code>javascript</code>	Obsolete in HTML5

8. Attribute Code Example

```
<script src="https://cdn.example.com/app.js" defer type="module" crossorigin="anonymous" integrity="sha384-abc"></script>
```

9. Real-World Case Study / Example

- **Case Study:** Included on every interactive web application to power JavaScript frameworks (React, Vue) and DOM logic.

```
<script src="bootstrap.bundle.min.js" defer></script>
```

10. Important Notes

Place external `<script>` links inside `<head>` with the `defer` attribute to prevent blocking HTML parser rendering.

11. Related Tags

Related to `<noscript>`, `<template>`.

12. Deprecated / Obsolete Status

Standard HTML5 Tag. Legacy `language="javascript"` attribute is obsolete in HTML5.

MODULE 8: CSE SYLLABUS THEORY & CONCEPTUAL REFERENCE

8.1 Exhaustive HTML Global Attributes Reference Table

Global attributes are attributes that can be used on **any HTML5 element**.

Attribute	Purpose / Function	Valid Values	Example Usage
id	Unique identifier for an element across the entire DOM tree. Used for CSS styling, JS DOM selection, and URL fragment anchors.	Unique string	<div id="header-bar">
class	Groups multiple elements under common class names for CSS styling and JS selection.	Space-separated class names	<p class="lead text-primary">
style	Applies inline CSS styling rules directly on the element.	CSS declarations string	<h1 style="color: blue;">
title	Provides advisory tooltip text shown when hovering mouse over element.	Tooltip string	<abbr title="HyperText Markup Language">HTML</abbr>
lang	Specifies the language code of the element's text content.	Language code (en , hi)	<p lang="fr">Bonjour</p>
dir	Specifies text directionality.	ltr , rtl , auto	<p dir="rtl">سلام</p>
hidden	Hides element visually from page display.	hidden , until-found	<div hidden>Secret</div>
tabindex	Controls keyboard Tab key navigation focus order.	Integer (0 , -1 , 1)	<div tabindex="0">
contenteditable	Allows user to edit element text content live.	true , false	<div contenteditable="true">
draggable	Specifies whether element is draggable via Drag & Drop API.	true , false , auto	
spellcheck	Controls browser spelling/grammar checking.	true , false	<textarea spellcheck="true">
translate	Hints whether text should be translated by translation tools.	yes , no	Brand
accesskey	Keyboard shortcut key trigger to focus element.	Single character	<button accesskey="s">
data-*	Custom data payload attribute for developer JavaScript storage.	String	<button data-user-id="1092">
role	WAI-ARIA role defining element semantic purpose for accessibility.	ARIA role string (button , navigation)	<div role="button">

8.2 Event Handler Attributes vs JavaScript Event Listeners

Event handler attributes execute inline JavaScript code when specific user interactions occur.

Common Inline Event Attributes Table

Event Attribute	Trigger Condition	Example Usage
onclick	Fires when element is clicked.	<button onclick="submitForm()">Submit</button>
onchange	Fires when input value is committed/changed.	<select onchange="updateCity(this.value)">
oninput	Fires instantly as user types in text box.	<input oninput="liveSearch(this.value)">
onsubmit	Fires when form submission occurs.	<form onsubmit="return validate()">
onload	Fires when element/page finishes loading.	<body onload="initApp()">

Why Modern Web Development Prefers addEventListener()

- 1. **Separation of Concerns (SoC):** Keeps HTML markup clean and moves JavaScript logic into modular .js files.
- 2. **Security & CSP:** Inline event handlers violate Content Security Policy (unsafe-inline) rules and expose sites to XSS security vulnerabilities.
- 3. **Multiple Event Binding:** Inline attributes allow only ONE handler assignment per event. addEventListener() allows attaching multiple independent listener functions.

```
<!-- DISCOURAGED (Inline Handler) -->
<button onclick="saveData()">Save</button>

<!-- RECOMMENDED (Modular JS Listener) -->
<button id="saveBtn">Save</button>
<script>
  document.getElementById("saveBtn").addEventListener("click", function() {
    console.log("Data Saved Securely");
  });
</script>
```

8.3 Void (Self-Closing) Elements vs Container Elements

- 1. **Container Elements (Paired Tags):** Have an opening tag (<tag>) and a closing tag (</tag>). They encapsulate inner text or child HTML elements. Examples: <div> , <p> , <h1> , <table> , <form> .
- 2. **Void Elements (Self-Closing Tags):** Cannot contain child text or inner HTML content. They do NOT have closing tags (</input> or cause syntax errors in HTML5). Examples:
 , <hr> , , <input> , <meta> , <link> , <source> .

8.4 Block-Level vs Inline Elements Comparison

Aspect / Feature	Block-Level Elements	Inline Elements
Default Display	<code>display: block;</code>	<code>display: inline;</code>
Line Break	Always starts on a new line ; forces newline after.	Fits inline within surrounding text flow.
Width	Occupies 100% full width of parent container.	Occupies ONLY as much width as content requires.
Margins & Padding	Respects all 4 sides (Top, Bottom, Left, Right).	Respects Left & Right; ignores Top & Bottom vertical height margins.
Key Examples	<code><div></code> , <code><p></code> , <code><h1>-<h6></code> , <code><section></code> , <code><article></code> , <code></code> , <code><table></code> , <code><form></code>	<code></code> , <code><a></code> , <code></code> , <code></code> , <code></code> , <code><code></code> , <code><label></code> , <code><input></code>

MODULE 9: COMPLETE CSE LAB ASSIGNMENT HTML5 PAGE

Below is a complete, fully functional HTML5 webpage designed according to standard 3rd Semester CSE Web Technologies lab curriculum guidelines.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="VTU CSE Web Technologies Lab Assignment 1">
  <title>VTU CSE Department | Student Registration Portal</title>

  <style>
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #f8fafc;
      color: #0f172a;
      line-height: 1.6;
    }
    header {
      background-color: #0f172a;
      color: white;
      text-align: center;
      padding: 1.5rem;
    }
    nav {
      background-color: #1e293b;
      padding: 0.75rem;
      text-align: center;
    }
    nav a {
      color: #38bdf8;
      margin: 0 15px;
      text-decoration: none;
      font-weight: bold;
    }
    .container {
      max-width: 1000px;
      margin: 20px auto;
      padding: 20px;
      background: white;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0,0,0,0.1);
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin: 15px 0;
    }
    table, th, td {
      border: 1px solid #cbd5e1;
    }
    th, td {
      padding: 10px;
      text-align: left;
    }
    th {
      background-color: #f1f5f9;
    }
    fieldset {
      border: 1px solid #cbd5e1;
      padding: 15px;
      border-radius: 6px;
      margin-bottom: 15px;
    }
    legend {
      font-weight: bold;
    }
    footer {
      background-color: #0f172a;
      color: white;
      text-align: center;
      padding: 1rem;
      margin-top: 30px;
    }
  </style>
</head>
<body>

  <!-- Document Header -->
  <header>
    <h1>Visvesvaraya Technological University</h1>
    <p>Department of Computer Science & Engineering</p>
  </header>

  <!-- Navigation Bar -->
  <nav aria-label="Main Navigation">
    <a href="#about">Course Overview</a>
    <a href="#schedule">Exam Schedule</a>
    <a href="#register">Student Registration</a>
  </nav>

  <!-- Main Content Container -->
```

```

<main class="container" role="main">

  <!-- About Section -->
  <section id="about">
    <article>
      <h2>Web Technologies (18CS52)</h2>
      <p>This lab course introduces students to <strong>HTML5</strong>, <em>CSS3</em>, JavaScript, and server-side web development.</p>
    </article>
  </section>

  <hr>

  <!-- Table Section -->
  <section id="schedule">
    <h2>Semester 3 Examination Timetable</h2>
    <table>
      <caption>Table 1: CSE Exam Schedule 2026</caption>
      <thead>
        <tr>
          <th scope="col">Subject Code</th>
          <th scope="col">Subject Name</th>
          <th scope="col" colspan="2">Date & Session</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <td>18CS51</td>
          <td>Management & Entrepreneurship</td>
          <td>01 Sept 2026</td>
          <td>Morning</td>
        </tr>
        <tr>
          <td>18CS52</td>
          <td>Web Technology & Applications</td>
          <td>04 Sept 2026</td>
          <td>Morning</td>
        </tr>
      </tbody>
    </table>
  </section>

  <hr>

  <!-- Registration Form Section -->
  <section id="register">
    <h2>Lab Assignment Submission Form</h2>
    <form action="/submit-assignment" method="POST" enctype="multipart/form-data" autocomplete="on">

      <fieldset>
        <legend>Student Personal Details</legend>

        <p>
          <label for="usn">USN Number:</label><br>
          <input type="text" id="usn" name="student_usn" placeholder="e.g. 1VU22CS001" required pattern="1[A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{3}">
        </p>

        <p>
          <label for="email">College Email:</label><br>
          <input type="email" id="email" name="student_email" placeholder="student@college.edu" required autocomplete="email">
        </p>

        <p>
          <label for="branch">Select Branch:</label><br>
          <select id="branch" name="branch" required>
            <option value="" disabled selected>-- Select Branch --</option>
            <option value="cse">Computer Science (CSE)</option>
            <option value="ise">Information Science (ISE)</option>
          </select>
        </p>
      </fieldset>

      <fieldset>
        <legend>Assignment File Submission</legend>

        <p>
          <label for="file">Upload Code (.pdf / .zip):</label><br>
          <input type="file" id="file" name="code_file" accept=".pdf,.zip" required>
        </p>

        <p>
          <label for="remarks">Remarks:</label><br>
          <textarea id="remarks" name="student_remarks" rows="3" cols="40" placeholder="Enter optional comments..." maxlength="500"></textarea>
        </p>
      </fieldset>

      <button type="submit">Submit Assignment</button>
      <button type="reset">Reset Form</button>
    </form>
  </section>

</main>

```

```
<!-- Document Footer -->
<footer>
  <p>&copy; 2026 VTU Department of Computer Science. All rights reserved.</p>
</footer>

</body>
</html>
```